

Usama Ikram

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RESEARCH INTERESTS

Machine learning and computer vision, with an emphasis on multimodal and foundation models and on the conditions under which such models remain reliable once deployed. My research so far has addressed the adaptation of large pretrained vision and vision-language models to settings where computation, latency and labelled domain data are all limited, using parameter-efficient methods and evaluating the resulting systems against the requirements of the application rather than benchmark accuracy alone. I am now interested in extending this work toward broader multimodal and foundation-model systems, including vision-language and language models, and toward healthcare and other applied domains where efficiency, reliability and practical usefulness are equally weighted.

EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

MS in Artificial Intelligence

Daegu, South Korea

Sep 2025 – expected Jun 2027

National University of Sciences and Technology (NUST)

BS in Electrical Engineering

Islamabad, Pakistan

2019 – 2023

MANUSCRIPTS UNDER REVIEW

All three manuscripts are first-authored and currently under peer review. None has been accepted or published.

1. **U. Ikram**, Y. Kim, and I. Moon. “Delay-Aware Deployability of Multimodal Foundation Models for Real-Time Human-Robot Interaction.” Submitted to *IEEE Transactions on Human-Machine Systems*. Under review.
2. **U. Ikram** and I. Moon. “Physics-Aware LoRA-Adapted Lightweight AI for Edge-Efficient Quantitative Holographic Cell Analysis.” Submitted to *Applied Physics B: Lasers and Optics*. Under review.
3. **U. Ikram** (corresponding author), A. Jose, F. C. Eyzaguirre, and A. Mac Cawley. “Determinants of Glycaemic Outcomes in Children and Adolescents with Type 1 Diabetes: A Systematic Review and Evidence Map.” Submitted to *Diabetologia*. PROSPERO CRD420251045872. Under review.

RESEARCH EXPERIENCE

Graduate Researcher, Department of Artificial Intelligence, DGIST

Daegu, South Korea

Sep 2025 – present

Deployability of multimodal foundation models in real-time interaction manuscript under review, *IEEE Transactions on Human-Machine Systems*

- Investigated why benchmark accuracy is insufficient to determine whether a multimodal model can be placed inside a wearable perception-in-the-loop system, in which inference delay acts as a transport lag on the closed loop.
- Formulated a control-theoretic deployability criterion: modelled the interaction loop as a PID-controlled mass-spring-damper plant with perception delay, applied a first-order Padé approximation to the delay term, and evaluated the resulting fourth-order characteristic polynomial under the Routh-Hurwitz criterion to obtain a delay-dependent stability bound of 37.77 ms.
- Adapted CLIP (ViT-B/32) with LoRA on a curated Conceptual Captions subset and benchmarked it against the unadapted backbone and a Frozen Prefix-LM across zero-shot, few-shot and linear-probe protocols on five datasets; LoRA preserved representation quality (linear-probe average 85.69% vs. 85.95%) while improving out-of-distribution transfer on remote-sensing imagery by 6.05%.
- Characterised latency, memory and energy for each configuration and defined a worst-case latency incorporating scheduling jitter and a deployment margin. Merged LoRA-CLIP achieved 17.92 ms worst-case latency, within both the nominal bound and the 18.76 ms bound that survives $\pm 50\%$ variation in biomechanical parameters; the Frozen Prefix-LM reached 442.02 ms end to end, exceeding the bound by $11.7\times$ because of autoregressive decoding rather than visual encoding.
- Validated the analysis in software-in-the-loop simulation, which remained bounded at 30 ms and diverged at 40 ms, and proposed a tiered deployment strategy assigning parameter-efficient models to high-rate perception and generative models to asynchronous reasoning. Latency and energy were measured on a desktop GPU; edge figures are reported as projections.

- Addressed a failure mode of conventional segmentation objectives in label-free cell imaging, where the predicted boundary determines a downstream biophysical measurement and pixel-overlap losses leave that measurement unconstrained.
- Designed a composite training objective combining region overlap with three domain-specific terms (contrast, boundary-gradient alignment and volume preservation), and established through component-wise ablation that the terms must be composed hierarchically: applied without a geometric anchor, the contrast term degraded boundary F1 to 0.386 and eliminated the rarest class.
- Reduced the resulting measurement error from 5.22% to 3.04% on EdgeSAM and from 15.52% to 3.01% on MobileNet-UNet relative to a Dice-only baseline, while attaining the best boundary fidelity of every configuration tested (0.953 and 0.970).
- Compared LoRA against full fine-tuning across three lightweight backbones. Full fine-tuning eliminated a diagnostically informative minority class in all three architectures, whereas LoRA at rank 8 retained all four classes in both attention-based backbones. As latency and memory were invariant across ranks, the low-rank subspace functions as a regulariser rather than a compute saving, which is the more consequential finding.
- Evaluated the framework on 1,974 individually segmented cells across an 11-timepoint, 47-day study, recovering the expected biological trajectory and agreeing with annotator-derived measurements at $r = 0.966$. Optimised deployment with ONNX FP16 compilation to 4.33 ms median latency (228 FPS) at 16.4 MB peak allocated memory on a desktop RTX A5000; embedded validation remains future work and is not claimed.

End-to-end lightweight holographic analysis *ongoing; recently initiated, no results yet*

- Developing a unified model that maps a raw hologram directly to both a reconstructed phase image and a cell segmentation map, replacing the conventional sequence of numerical reconstruction followed by independent segmentation, and derives per-cell morphology, projected area, optical volume and dry mass from those outputs.
- Comparing two acquisition configurations, raw off-axis and raw in-line Gabor holograms, across reconstruction accuracy, segmentation quality, morphology classification, measurement accuracy and computational cost, in order to establish what the simpler and cheaper configuration actually costs at the level of the quantities used downstream.
- Extending the earlier loss framework to a joint objective over reconstruction, segmentation and measurement consistency.
- Responsible for network architecture, loss design, dataset organisation and the experimental comparison protocol; hologram datasets are provided by a collaborator. Target for an initial manuscript: September 2026.

Evidence synthesis in paediatric type 1 diabetes *manuscript under review, Diabetologia; with Pontificia Universidad Católica de Chile and University of Limerick*

- Led a PROSPERO-registered systematic review of 63 quantitative studies selected from 5,406 records across four databases, screened in duplicate, appraised with the Mixed Methods Appraisal Tool and reported to PRISMA 2020.
- Constructed the PEARL framework: 47 determinants organised into 12 sub-domains and five domains under four stated assignment rules, with a counted weight of evidence attached to each category, which comparable frameworks do not provide.
- Identified an inverse relationship between evidence volume and consistency across domains, with the uniformly significant psychological literature (26 of 26 associations) more plausibly explained by selective reporting than by larger effects, and set out the reporting practices needed to correct it.

PROFESSIONAL EXPERIENCE

National Aerospace Science & Technology Park (NASTP)

AI & ML Scientist

Rawalpindi, Pakistan

Jul 2025 – Sep 2025

- Developed object detection and region-of-interest segmentation pipelines for surveillance imagery and deployed them on RK3588 SoC hardware, establishing the practical cost of on-device inference for low-power camera prototypes.
- Implemented and benchmarked autoencoder architectures for representation learning over multimodal sensor data.
- Designed an LLM-based candidate screening system that generates a technical assessment from each applicant's CV and flags inconsistencies in employment history.

DeepChain

AI Research Engineer

Islamabad, Pakistan

Jul 2023 – Mar 2025

- Built and trained convolutional, recurrent and autoencoder models for client deployments in supply chain automation and industrial monitoring.

- Diagnosed data loss between field sensors and collection terminals and rebuilt the ingestion pipeline, so that models were trained on the signal the sensors produced rather than on gap-filled records.
- Conducted on-site data collection to obtain reliably labelled ground truth, and carried results through exploratory analysis, feature engineering and technical reporting to clients.

National Cyber Security Auditing and Evaluation Lab (NCSAEL)

Rawalpindi, Pakistan

Frontend Web Development Intern

Jul 2021 – Oct 2021

- Implemented responsive web interfaces and resolved front-end defects alongside the development team.

SELECTED EARLIER PROJECTS

- **Explainable ensemble for knee osteoarthritis grading** | [code](#). Automated Kellgren-Lawrence grading from OAI knee radiographs using a heterogeneous ensemble of CNN backbones combined by a mixed voting rule, with Grad-CAM and LIME attribution used to verify that predictions rested on joint space narrowing rather than on imaging artefacts.
- **Early detection of Alzheimer’s disease on embedded hardware** | [code](#). CNN classification of ADNI imaging combined with ridge regression for physiological score prediction and an SVM stage for identifying likely progression, deployed on a Jetson Nano to assess whether the pipeline fits an embedded power budget.
- **Predictive maintenance from industrial IoT sensor data**. Standardised accelerometer vibration readings across injection moulding machines and trained a neural classifier to separate healthy from degrading operating states, enabling maintenance to be scheduled before failure.
- **Indoor machine positioning by trilateration**. Measured Bluetooth signal strength from deployed IoT terminals on a plant floor and recovered machine coordinates by trilateration for equipment tracking.

TECHNICAL SKILLS

Languages	Python, C++ , MATLAB, SQL, W TeX, HTML/CSS
Machine learning	PyTorch, TensorFlow, Hugging Face, PEFT/LoRA, ONNX Runtime, mixed-precision training, CNNs, transformers, autoencoders, recurrent models
Vision & multimodal	Vision-language models (CLIP, prefix-tuned LMs), semantic and instance segmentation, object detection, biomedical and holographic image analysis, OpenCV
Deployment	Latency, memory and energy profiling, graph compilation and quantisation, embedded inference (RK3588, Jetson Nano), Docker, Git, Flask
Analysis	Control-theoretic stability analysis, domain-constrained loss design, statistical analysis, systematic review methodology (PRISMA 2020, MMAT)
Spoken	English (professional), Urdu (native)

Last updated: August 2026.